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An Innovative Novel Transformer Model and Datasets for Safeguarding Religious Sensitivities in Online Social Platforms

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Abstract

Nowadays, blasphemy, inter-religious hatred, and intolerance toward other people's faith become increasing problems in different social media. To effectively tackle these problems, it is necessary to develop efficient techniques for automated detection. With the increasing concern about hate speech on the internet, this research study focuses on creating and testing advanced models to identify and categorize texts related to religious hatred. Leveraging two meticulously curated datasets, the first comprising binary classification tasks to discern hate speech directed towards religion and the second featuring a nuanced five-class categorization including Blasphemy, Ethnic Hatred, Ethnic Group Hatred, Undetermined, and Non-Hatred. The main focus of the research is examining the effectiveness of different established machine learning techniques, deep learning methods, and pre-trained transformer models. Notably, the study introduces hatebnBERT, a novel BERT-based model tailored to the Bangla language's specific nuances and linguistic intricacies specifically built for hate speech detection. Through rigorous evaluation and comparative analysis, the results demonstrate the superior accuracy and precision of hatebnBERT, achieving an accuracy of 98.8% in the binary-class dataset and 98.6% in the multiclass dataset, in effectively identifying and classifying various forms of hate speech within the realm of religious discourse. The outcomes underscore the significance of leveraging specialized language models, such as hatebnBERT, to combat the proliferation of abusive speech and promote a more courteous and inclusive online social media environment within the Bangla-speaking community, considering the potentially devastating consequences of unchecked religious hatred propagation. Insert here your abstract text.

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Keywords: Hate speech classification ; NLP ; Bangla text dataset ; Transformers ; Religion.

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1. Introduction

Natural Language Processing merges Linguistics and AI to help machines understand human language. It includes Natural Language Generation and Understanding and can improve our comprehension of language [1]. Detecting and classifying hate speech is a significant research in natural language processing(NLP), which is crucial. This task can address one of the challenges in the field [2]. Blasphemy is disrespect to God or holy things. Heresy is a belief that goes against the official religion. Religious intolerance is a lack of acceptance of others' beliefs. The rise of blasphemy, religious intolerance, and disrespect towards people's beliefs expressed through social media and online forums can have severe consequences on individuals and communities. To ensure a safe and inclusive online environment, it is crucial to predict and prevent hate speech in different languages. detecting hate speech in high-resource languages, like as English, has been extensively researched, as evidenced by [3]. However, for low-resource languages like Bengali, this is not the case. Bengali is the seventh most spoken language in the world and has a significant online community. Due to the absence of annotated data and linguistic resources, it is a significant challenge to recognize hate speech in this language [4]. Besides, rumors [5] hate speech towards any group of religion is a sensitive issue that can lead to riots and devastating situations[6]. By enabling the identification and removal of hateful content from online platforms, a safer environment for users can be created. Hate speech in the domain of religion has become a major concern as social media platforms continue to gain more influence in recent times. The need for an effective hate speech detection system, especially in the religious domain, has become more pressing than ever before. It is important to develop such a methodology that can predict and flag hateful content in the Bangla language to promote safe and respectful communication online. Though there are several studies[7] have been conducted on hate speech detection and classification in the Bangla language, no research has been done yet on detecting hate speech related to religious perspectives. The objective of this study is to address the lack of datasets related to hate speech in the religious domain. To achieve this goal, the research will involve exploring advanced models of classification and developing a classification model that is tailored to the new dataset and its specific purpose. The objectives of this research are:

- To create a specialized dataset in the Bangla language for religious hate speech, including binary classification tasks and a five-class categorization (Blasphemy, Ethnic Hatred, Indigenous Group Hatred, Undetermined, and Non-Hatred), to aid research and model evaluation.
- To detect and classify religiously motivated hate speech in Bangla text, we curated a dataset and evaluated many machine learning and deep learning models, including traditional classifiers, deep learning architectures, and transformer models.
- To introduce hatebnBERT, a specialized BERT-based model designed to detect and classify religiously motivated hate speech in Bangla text.
- To present insights for detecting and mitigating religious hatred in the Bangla-speaking community, using advanced language models to promote inclusive and respectful online discourse in the religious domain.

2. Related Work

The identification and elimination of hate speech on various online platforms has become an important area of research in recent years. The negative impact of expressing hateful sentiments on social media channels has been a driving force behind this research. As a result, the ability to automatically identify hate speech is becoming increasingly necessary[8]. Distinguishing between text messages containing hate speech and those that do not is typically considered a straightforward task. Various machine learning algorithms have been employed in English datasets to categorize abusive content on social media. [9] A recent study proposed an encoder-decoder model for classifying Bengali comments, employing 1D convolutional layers and attention-based decoders, achieving an accuracy of 77% in identifying seven distinct hate speech categories. The study compared various feature engineering strategies and found that the SVM algorithm combined with bigram features had the highest accuracy at 79%. These findings provide insights for further investigation into automatic text classification. The use of Deep Learning has gained popularity in detecting hate speech in recent times. Several research studies have been published that demonstrate the superiority of

neural networks over state-of-the-art machine learning classifiers [10].In this study [11], researchers proposed a Deep Convolutional Neural Network (DCNN) that utilizes tweet text and pre-trained GloVe embedding to extract semantic features. They found that the proposed DCNN model outperformed many baseline Machine Learning classifiers and Long Short-Term Memory (LSTM) models with an F1-score of 0.92 and a misclassification rate of only about 2%. This study highlights the potential of using DCNN models for text classification tasks. Amid the surge in online activity during the COVID-19 pandemic, the prevalence of offensive Bengali and Banglish content on social media has highlighted the need for effective detection mechanisms.[12] This study introduces L-Boost, an advanced algorithm combining AdaBoost and LSTM models with BERT, achieving a remarkable 95.11% accuracy, surpassing traditional baseline classifiers for identifying offensive text. [13]This research delves into the use of Transformer-based Machine Learning (ML) models, such as mBERT and XLMR, to identify hate speech in English and Indo-Aryan languages. The models developed by the "Super Mario" team performed exceptionally well in various categories at FIRE 2021. They secured the 2nd position in the Code-Mixed Data set (Macro F1 score: 0.7107) and Hindi two-class classification (Macro F1 score: 0.7797), the 4th position in English four-class categorization (Macro F1 score: 0.8006), and the 12th position in English two-class categorization (Macro F1 score: 0.6447). [14]In order to identify and eliminate hateful comments, this research concentrates on the Urdu language and creates a list of offensive vocabulary along with a dataset of 10,526 Urdu tweets that have been annotated. By utilizing transfer learning with pre-trained FastText Urdu word embeddings and multilingual BERT embeddings, the study demonstrates encouraging F1 scores that outperform various baseline models. The study showcases the dangers of hate speech and introduces an automated system for detecting Bangla hate speech using NLP and deep learning techniques.[15]. By merging diverse datasets and implementing preprocessing methods, the research demonstrates the superior performance of the GRU and Attention techniques, achieving accuracies of 98.87% and 98%, respectively, in identifying Bangla hate speech. The study addresses the global challenge of cyberbullying, introducing a robust detection system for Bangla that achieves a 98% accuracy in threat detection and 90% accuracy in identifying sarcastic comments. By incorporating diverse textual features, including social profiles and contextual embeddings, the method[16] outperforms existing approaches, showcasing its efficacy in combating cyberbullying. Previous works have introduced datasets on different languages, machine learning, and deep learning approaches. However, This study presents a unique dataset and a transformer model called hatebnBERT designed to detect religiously motivated hate speech in the Bangla language with high accuracy. It provides a comprehensive analysis of hate speech detection in the Bangla religious domain, using a detailed multi-class classification system and evaluating various machine learning models.

3. Dataset Description

As part of a study on offensive speech from a religious perspective, two new datasets for Bengali text were created. An overview of the protocol used to generate the datasets is presented in Figure 1. It can be observed from the figure that the protocol is composed of four major steps: data collecting, cleansing, annotations rules development, and data annotation. A group of three experts gathered data from social media platforms including Twitter, YouTube, and Facebook. They developed rules with high inter-annotator agreement (0.853) through multiple rounds of comparison and discussion, using the Kappa statistic.

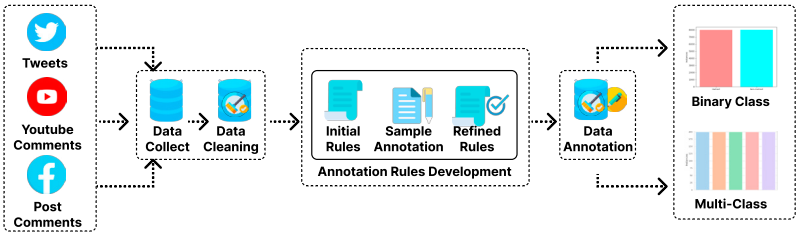
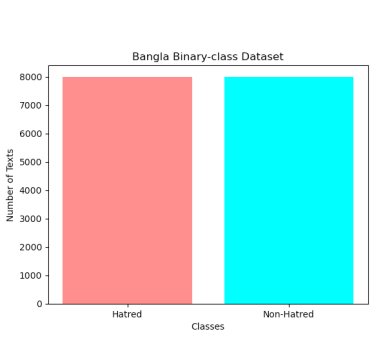
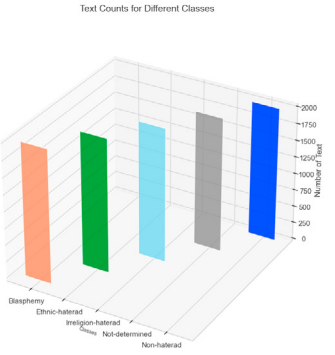


Fig. 1: Steps involved in creating the Bangla(Binary and Multiclass) datasets social media platform posts and comments

3.1. Binary-Dataset

The team analyzed content from groups, celebrity and local Bengali news pages, roasting videos, political news posts, and funny posts between January 1, 2022, and July 30, 2023. A total of 150,000 posts and comments were collected from these platforms and filtered to find 9,000 posts containing Bengali profanity to increase the chances of finding offensive content on Religion. To ensure class balance, we also identified 9,000 posts from the original dataset that did not contain offensive or hateful content. Initially, 18000 texts were labeled as hatred or non-hatred. After manual scrutiny, 16000 texts were kept for binary classification by removing noise and irrelevant elements. Figure 2a shows the binary class dataset text distribution.







(a)Data Distribution of Binary-class Dataset
Fig. 2 : (b)Banglamulti-classdataset.
(c) Wordcloud of frequent Bangla words addressed in hated texts

3.2. Multi-class Dataset

The multi-class dataset was created based on the specified rules, resulting in five classes: Blasphemy, Ethnic-Hatred, and Irreligion-Hatred, with additional categories of not-determined and Non-Hatred.Figure 2b shows the multi-class dataset distribution. **Blasphemy:**Texts were annotated as Blasphemy when any comment is made with the intention of hurting, threatening, or provoking someone, specifically in relation to their faith or belief. Insulting or expressing contempt for any deity, God, or demonstrating irreverence towards anything considered sacred or inviolable by a particular religion or group of people is included, as well as sarcasm regarding religion or faith.**Ethnic-Hatred:** Texts were categorized as Ethnic Hatred if they involved interfaith attacks, comparisons of religious groups to animals, blaming one faith for actions against others, cursing groups based on religion, demanding rights refusal, or accusing religious groups of disloyalty to the state. **Irreligion-hatred:** Irreligion encompasses various positions, including scepticism, agnosticism, rationalism, atheism, and secularism. texts annotated as Irreligion-hatred were directed towards this person in groups. **Not-determined:** Texts were annotated as Not-determined if texts included hateful words without a clear target community based on faith, sect, or ethnicity. Also included were insults towards individuals or groups based on caste or religion without commenting on their belief system, as well as hateful expressions towards communities based on characteristics other than faith, sect, or ethnicity. Finally, all nasty comments that don't fall into the categories of Blasphemy, Ethnic-hatred, and Irreligion-hatred are also considered Not-determined. **Not-hatred:** These were the texts that didn't contain any hateful or abusive words and were not detected as any of the four classes(Blasphemy, Ethnic-Hatred, Irreligion-hatred, Not-determined). The Multiclass dataset is shown in the table 3. The wordcloud displays the entities that are targeted or attacked by frequent use of Bangla hatred and abusive language. 2c.

4. Proposed Methodology

The methodology of this research began with collecting extensive data from various sources and annotating it meticulously to ensure accurate labeling, which is discussed. The data was preprocessed with multiple steps such as

Texts	label
তদের আগাকাডার ধর্ম হল পতিতার ধর্ম	Ethnic hatred
এই শালা মালাউনের দাদাল দুই	Ethnic hatred
মা বোনদের বল শিবলিঙ্গের সাথে সেক্স না করে আমাদের লিংগের সাথে সেক্স করতে বল	Blasphemy
আরে মালাউনের বাচ্চা ইসলামের মজা তোরা বুঝবিনা	Ethnic hatred
শালা মূর্তি পূজারী, শিবের ধনেরর পূজারি	Blasphemy
যারা আল্লাহ ধর্ম ইসলাম নালে না তাদের হত্যা করা জায়েজ	Irreligion-hatred
আমি দুর্গা জুতো ছুড়ে মারব, দুর্গার হাতে কি ট্যাটু ছিল?	Blasphemy
উপজাতি খবিশ গুলা যা পায় তাই খাই	Ethnic hatred
নাস্তিক মাদারচোদ খানকির পোলা ইসলাম নিয়ে কথা বলস	Irreligion-hatred
তগ জাতটাই খারাপ খানকির মাগির দল	Not-Determined
কাওমি হুজুর গুলা এক একটা ধর্ষক	Ethnic hatred
ধর্ম নিয়ে বাড়াবাড়ি কোন সুফল বয়ে আনবে না	Not-hatred
তর আল্লাকে কি এখন ঘুমায়	Blasphemy

Fig. 3: Sample texts of multi-class dataset

data cleaning, removing stop words, URLs, and emojis, and tokenizing. Stop word removal to ensure the quality and integrity of the dataset. It was further divided into distinct training and testing subsets, respectively, in 80% and 20%. The study involved training a variety of models, including classical machine learning models such as Random Forest(RF), Support Vector Machines (SVM), Naïve Bayes(NB), and Logistic Regression (LR), along with deep learning models, such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) and Bidirectional long-short term memory(Bi-LSTM). The research also delved into applying advanced transformer models, namely multi-lingual Bidirectional Encoder Representations from Transformers mBERT, BanglaBERT, and the proposed model hatebn-BERT, to explore their effectiveness in language processing tasks. The models were evaluated comprehensively, using precision, recall, F1 score, and accuracy metrics to determine their optimal performance. The workflow of the research methodology is shown in Figure 4.

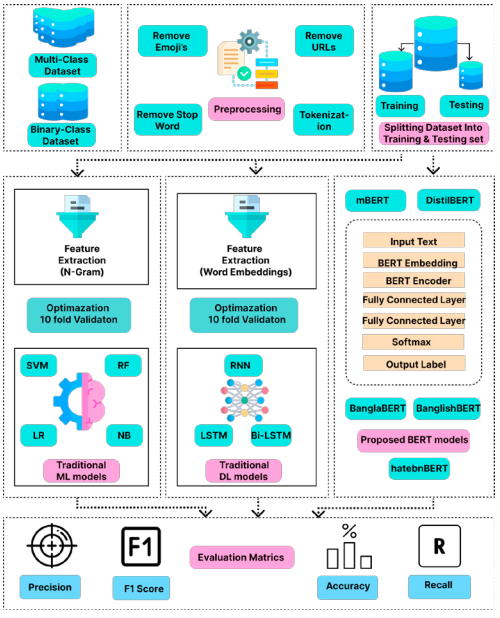


Fig. 4: Proposed workflow architectural analysis of this study

4.1. Machine Learning Models

The study utilized various machine learning models, including RF, SVM, Naïve Bayes, and Logistic Regression. SVM is well-known for its effectiveness in high-dimensional spaces, while RF is preferred for its ability to handle large datasets. NB is known for its simplicity and operates based on the assumption of feature independence. LR, a widely used baseline model, is chosen for its interpretability and efficiency in binary classification tasks. By combining these models, the research aims to thoroughly evaluate their performance in detecting and classifying hate speech within the Bangla religious domain.

4.2. Deep Learning Models

The investigation employed advanced deep learning techniques, such as Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (Bi-LSTM), and Recurrent Neural Networks (RNN). LSTM has been specifically designed to capture extended dependencies in ordered data, making it an appropriate choice for natural language processing-related tasks. An enhanced form of LSTM, known as Bi-LSTM, is capable of processing data in forward and backward directions simultaneously. This unique approach allows it to effectively capture contextual information from both previous and subsequent data points. On the other hand, RNN is particularly advantageous in tasks that involve the analysis of ordered information, such as NLP and time series prediction. The research aims to explore the efficacy of these models in discerning the complex patterns of hate speech within the Bangla religious domain.

4.3. Tranformer Models

mBERT[17] (multilingual BERT) and BanglaBERT[18] are both advanced transformer[19] models used in natural language processing tasks. The BERT model's multi-lingual variant, mBERT, has undergone training to process text in several languages simultaneously, enhancing its capacity to comprehend and analyze diverse languages. This model's versatility makes it valuable for tasks involving multilingual text analysis and classification. On the other hand, BanglaBERT is specifically fine-tuned and optimized for the Bangla language, allowing it to capture the intricacies and nuances of the language more effectively. With a focus on Bangla-specific linguistic features and patterns, BanglaBERT demonstrates superior performance in tasks related to Bangla text analysis and processing. The research aims to investigate the applicability of mBERT and BanglaBERT in detecting and categorizing hate speech in the Bangla religious domain. This will lead to a refined and precise comprehension of hate speech within the context of the Bangla language. Table 1 depicts the overview of the transformer models.

Table 1: Transformer Models Overview

Model	Type	Trained	Layer	Hidden	Head	Parameters
mBERT	BERT-Base, Multilingual Cased	Trained on cased text in the top 104 languages with the largest Wikipedias	12	768	12	110M
BanglaBERT	BanglaBERT (small)	BERT-based monolingual model pre-trained in Bangla Language. 27.5 GB of Bangla data used for pretraining	12	256	4	110M

4.4. hatebnBERT

To develop the hatebnBERT model for the Bengali language, the team followed a detailed process outlined in Figure 3. Initially, they gathered a significant corpus of offensive language written in Bengali. However, due to the limited availability of existing data, they translated a total of 12 English and Hindi offensive datasets (comprising 137,000 sentences in total) to Bengali utilizing the Google API. Furthermore, the team incorporated 22367 religiously hateful texts obtained from various social media platforms. Finally, these offensive sentences were used to retrain the existing BengaliBERT(sarker) model. The BanglaBERT(sarker)[20] model is a highly advanced language model for the Bengali language. It has been meticulously trained on a vast amount of data from Bengali Wikipedia pages, totaling 18.6 GB. The model has been put through a strict training process to ensure maximum accuracy and reliability. The process involved 1 million training steps and 3 billion tokens (24B characters) of Bengali text obtained from different online platforms like news, discussion forums, and internet crawls. We have improved the model by training it again with 1,59,367 texts from the offensive corpus. The total number of tokens in these texts is 40,309,341. To achieve this, we retrained the model using the Masked Language Model, for 10 epochs, with 64 samples per batch. Each sample contained up to 512 sentence tokens. The Adam optimizer with a learning rate of 5e-5 was used to optimize the Neural Network algorithm. The building architecture of hatebnBERT is shown in Figure 5.

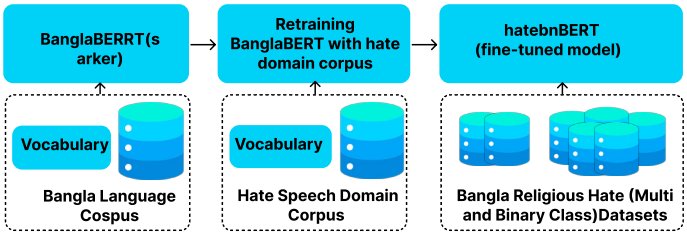


Fig. 5: Architecture of hatebnBERT

5. Experimental Result Analysis

To measure the effectiveness of these approaches, we take a closer look at their accuracy, precision, recall, and F1 score. These are the key metrics that help us determine how well these methods perform. One of the crucial assessment metric factors[21] is True Positive. It refers to the scenario when both the expected and actual numbers are positive.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

To calculate the precision metric[22], the total number of classified positive samples are divided by the number of correctly classified positive instances. The formula to compute this metric is expressed as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Recall is a metric in classification that measures true positives correctly identified from all actual positives. It shows the model's ability to detect all positive instances.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

The F1 score is a combination of precision and recall to measure a model's overall performance. It is calculated using the harmonic mean, which gives more weight to low values.

$$\text{F1-score} = \frac{2 \times Recall\ times\ Precision}{Recall + Precision} \tag{4}$$

Accuracy is important for classification tasks. Precision measures positive identification and reduces false positives. The F1 score is a balanced statistic that considers recall and accuracy, especially useful for unbalanced datasets. F1 score evaluates a model's performance by balancing accuracy and recall. A higher score indicates better accuracy and recall, while a lower score suggests an imbalance between the two. The methods' performance is validated using k-fold cross-validation. The data is split into ten folds to consider performance variations. The analysis focuses on the average k-fold accuracy and standard deviation.

5.1. Performance Analysis

The performance of all eleven models, including classical machine learning, Deep learning, transformer models, and finally, the proposed hatebnBERT models, is shown in Table 2.

5.2. Confusion Matrices of hatebnBERT Model

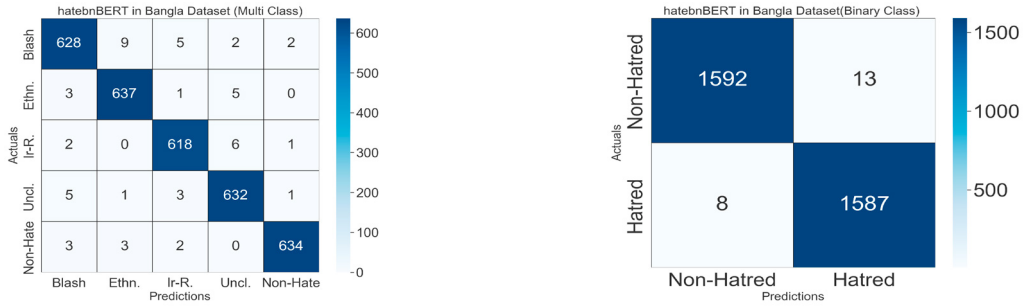
A summary of predicted and actual class labels that help evaluate a classification model's performance using true positive, true negative, false positive, and false negative values is known as a confusion matrix. The confusion matrices of the proposed hatebnBERT model for both Binary and Multi-class Bangla text datasets are depicted in Figure 6.

Table 2: Performance analysis of the models in both binary and multi-class Bangla Text Dataset

Model	**Accuracy**		**Metric**	**Binary-Class**		**Multi-Class**				
	Binary	**Multi**		**Hate.**	**Non-Hate.**	**Blash.**	**Ethn.**	**Ir-R.**	**Uncl.**	**Non-Hate.**
			P	0.50	0.63	0.38	0.94	0.97	0.96	0.49
NB	0.534	0.502	R	0.83	0.43	0.99	0.26	0.28	0.39	0.81
			F1	0.62	0.63	0.54	0.41	0.14	0.56	0.61
			P	0.63	0.55	0.42	0.64	0.52	0.84	0.64
LR	0.589	0.574	R	0.43	0.72	0.91	0.39	0.29	0.69	0.58
			F1	0.51	0.62	0.57	0.48	0.37	0.76	0.61
			P	0.59	0.63	0.50	0.51	0.51	0.84	0.68
RF	0.614	0.581	R	0.63	0.59	0.84	0.49	0.28	0.66	0.59
			F1	0.61	0.61	0.64	0.50	0.37	0.74	0.64
			P	0.62	0.62	0.75	0.79	0.42	0.81	0.58
SVM	0.622	0.612	R	0.68	0.56	0.47	0.42	0.68	0.62	0.82
			F1	0.65	0.58	0.57	0.55	0.52	0.70	0.68
			P	0.59	0.51	0.41	0.89	0.74	0.70	0.61
RNN	0.551	0.545	R	0.54	0.53	0.96	0.26	0.07	0.66	0.66
			F1	0.61	0.52	0.58	0.41	0.12	0.68	0.63
			P	0.71	0.59	0.76	0.79	0.37	0.72	0.62
LSTM	0.623	0.594	R	0.60	0.58	0.48	0.45	0.63	0.67	0.75
			F1	0.64	0.62	0.59	0.57	0.46	0.70	0.68
			P	0.72	0.68	0.91	0.77	0.35	0.93	0.96
Bi-LSTM	0.701	0.693	R	0.73	0.70	0.50	0.97	0.62	0.94	0.69
			F1	0.71	0.68	0.64	0.85	0.45	0.79	0.72
			P	0.92	0.94	0.94	0.95	0.91	0.93	0.96
mBERT	0.941	0.934	R	0.96	0.95	0.88	0.98	0.97	0.98	0.87
			F1	0.95	0.92	0.91	0.97	0.94	0.95	0.91
			P	0.96	0.97	0.98	0.98	0.98	0.98	0.99
Bangla-BERT	0.968	0.961	R	0.97	0.97	0.97	0.98	0.99	0.98	0.99
			F1	0.94	0.96	0.98	0.97	0.99	0.99	0.98
			P	**0.99**	**0.99**	**0.98**	**0.98**	**0.98**	**0.98**	**0.99**
hatebn-BERT	**0.988**	**0.986**	**R**	**0.98**	**0.99**	**0.97**	**0.98**	**0.99**	**0.98**	**0.99**
			F1	**0.99**	**0.98**	**0.98**	**0.97**	**0.99**	**0.99**	**0.98**

6. Result & Discussion

This research study explores the performance of various traditional machine learning classifiers, deep learning models, and pre-trained transformer models, including mBERT, BanglaBERT, and the proposed model, hatebnBERT,



(a) hatebnBERT(multiclass) (b) hatebnBERT(binary-class)

Fig. 6: Confusion Matrix of bnhateBERT an in Bangla binary multiclass Dataset

across binary and multiclass classification tasks. Two distinct datasets, one binary-class and the other multiclass, consisting of Bangla text, were employed for evaluation.

6.1. Performance of Traditional Machine Learning, Deep Learning, and Pre-trained Transformer Models

In this study, four different algorithms were tested for their performance in binary-class and multiclass datasets. Naïve Bayes showed an accuracy of 53.4% and 50.2%, respectively, with moderate precision values. LR achieved an accuracy of 58.% and 57.4% respectively, with better precision and recall balance. Random Forest had an accuracy of 61.4% and 58.1%, respectively, with moderate precision values. SVM exhibited an accuracy of 62.2% and 61.2%, respectively, with well-balanced precision scores for most classes. The Recurrent Neural Network (RNN) achieved an accuracy of 55.1% in the binary-class dataset and 54.5% in the multiclass dataset. Although the accuracy is relatively low, the RNN model showed a better balance between precision and recall in the binary-class context. However, the model's performance was inconsistent across different classes in the multiclass scenario. The Long Short-Term Memory (LSTM) model displayed an accuracy of 62.3% in the binary-class dataset and 59.4% in the multiclass dataset. The LSTM model showed balanced precision and recall values across most classes in both binary and multiclass tasks. The Bidirectional LSTM (Bi-LSTM) model demonstrated an accuracy of 70.1% in the binary-class dataset and 69.3% in the multiclass dataset. The Bi-LSTM model exhibited the best performance among deep learning models, with high precision and recall values across binary and multiclass scenarios. The mBERT model showed outstanding performance with an accuracy rate of 94.1% in the binary-class dataset and 93.4% in the multiclass dataset. It consistently demonstrated high precision and recall values across all classes in both datasets. Similarly, BanglaBERT also demonstrated exceptional performance with an accuracy rate of 96.8% in the binary-class dataset and 96.1% in the multiclass dataset. It consistently exhibited high precision and recall values across all classes in both contexts.

6.2. Proposed Model: hatebnBERT

hatebnBERT, the model proposed in this research, outperformed all other models with an accuracy of 98.8% in the binary-class dataset and 98.6% in the multiclass dataset. The model consistently exhibited high precision and recall values across all classes in both datasets, indicating its effectiveness in detecting and classifying the religion domain-based abusive content in Bangla text. Overall, the results of this study indicate the superiority of pre-trained transformer models, particularly BanglaBERT and the proposed model, hatebnBERT, in achieving high accuracy and balanced performance in both binary-class and multiclass classification tasks. These findings suggest the potential for these models to be deployed effectively in real-world applications for Bangla text classification tasks. Furthermore, the research highlight

7. Conclusion & Future Work

This study demonstrates the effectiveness of transformer-based models, especially hatebnBERT, in accurately identifying and classifying abusive and non-abusive Bangla text within the domain of religion. The superior performance of hatebnBERT, along with the promising outcomes of mBERT and BanglaBERT, highlights the potential of advanced natural language processing techniques in promoting a safer online environment. In the future, research should focus on fine-tuning hatebnBERT, expanding the dataset to encompass diverse religious text samples, contextual analysis of identified hate speech, integration of multimodal analysis techniques, and in-depth investigations into the ethical and societal implications of hate speech detection within religious contexts. This will promote developing more robust and ethically sound AI-driven solutions for content moderation in Bangla texts.

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